钱家伟 / Jiawei Qian

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PROFESSIONAL EXPERIENCE

AI Agent Engineer, Meituan

Sep 2025 – Present

End-to-end owner of AI Browser core intelligence and experience, responsible for agent quality iteration, deep user-scenario definition, core case design, and next-generation product capability evolution. Downward scope covers the Agent Harness self-iteration system, agent runtime architecture, and training flywheel; upward scope defines the product form, capability boundary, and experience baseline of Browser x Agent across browser-use / computer-use scenarios.

Methodology

- Designed the Agent Loop through a control-theoretic lens: treated the LLM policy as the plant, the harness as the controller, and evaluation / semantic observer as feedback sensors; explicitly separated setpoint, state, observation, and actuation to reduce oscillation and divergence in long-horizon tasks.
- Reframed agent-quality iteration as a training-style process: benchmark score becomes the objective, structured attribution from auto-analysis becomes the loss, and context structure, memory organization, Agent Skills, and system prompts become the updatable parameter space for gradient-like iteration.

Agent Harness Self-Iteration Pipeline (Core Direction)

- Designed and shipped the closed-loop auto-evaluation -> auto-analysis -> auto-iteration system: a higher-order agent controller evaluates sub-agent runs, attributes failures, decides iteration plans, and applies updates, upgrading agent improvement from human-driven tuning to system-driven iteration.
- Decomposed the full adjustable surface of agent quality into independently optimizable dimensions: context structure and denoising, tool-space definition and composition, system prompt engineering, Agent Skill injection and orchestration, and GUI Agent vs CLI Agent vs hybrid-mode routing.
- Built automatic Skill / Playbook distillation and injection: critical paths, UI elements, and action sequences are traced from execution trajectories, converted into reusable Agent Skills, injected at test time, and reused for self-evolution as a skill-level iteration flywheel.
- Designed the evaluation layer with rule-based checks and LLM-as-judge rubrics, combining task completion, trajectory efficiency, and behavioral compliance into structured attribution signals for auto-analysis rather than a simple pass rate.
- Designed the analysis and decision layer that maps evaluation signals to bottleneck dimensions such as missing context, tool gaps, prompt ambiguity, or skill gaps, then generates executable iteration proposals and applies them back into the harness.

User Case Mining & Agent Capability Definition

- User case mining & benchmark construction: mined measurable and regression-ready high-value task suites from real user journeys, including long-horizon information gathering, cross-tab / cross-site coordination, form and transaction operations, and complex web understanding, using them as agent setpoints and benchmark cases.
- Agent form & capability definition: defined the next-generation Browser x Agent interaction pattern, deep-scenario boundary, and experience baseline; mapped model upgrades, harness evolution, skill distillation, and context / memory mechanisms to concrete capability stages from usable to stable, predictable, and scalable.

Browser-use & Computer-use Agent Runtime

- Designed and iterated a ReAct-loop-based agent runtime covering state management, observation abstraction, tool-calling protocol, and failure recovery, improving execution stability for long-horizon browser tasks.
- Decoupled soft optimization from hard constraints by introducing the State Tracker and Semantic Observer, shifting robustness from model-dependent behavior to architecture-backed guarantees and reducing logical oscillation in complex tasks.

Training Flywheel

- Explored an SFT data flywheel based on real expert trajectories, built lightweight-model alternatives for selected vertical tasks, and validated the replacement potential of 7B / 30B models in high-frequency scenarios to reduce token cost and end-to-end latency.

Software Engineer, Microsoft

May 2024 – Sep 2025

负责 Agent 驱动的数据分析系统、LLM evaluation pipeline 与大规模数据基础设施建设，持续覆盖 agent workflow、代码生成、memory/context optimization 以及 cost/performance optimization。

Agent Workflow, Code Generation & Memory Optimization

- 推动分析与研发流程从人工执行转向 Agent + code generation + validation 驱动的 AI-native workflow，通过任务拆解、自动验证与迭代重试减少人工操作并提升交付效率。
- 基于 GPT-4o、LangGraph 与 OpenManus / MetaGPT 开发数据分析 Agent，打通 User Query -> Agent -> 数据分析与可视化的核心流程，支持结合 RAG、领域知识图谱和内部分析工具完成自动化分析与报告生成。
- 迭代开发 SQL -> DataFrame 代码生成与 sandbox validation -> re-generate 闭环，提升分析代码生成的可执行性与可靠性。
- 基于 MemGPT 思路对 external knowledge、chat history 和 tool feedback 进行 context 分层管理，减少 30% token usage per task，并提升多轮分析任务表现。

LLM Evaluation & Data Flywheel Infrastructure

- 基于 GPT-4o、o1、DeepSeek V3 / R1 等模型构建 Azure ML 评估管道，结合用户信号生成长期画像，并对 MSN / Bing 等场景中的 ranking 与 feed relevance 进行自动化评估和监控，处理规模约 416M tokens/day。
- 围绕 embedding 与文本处理链路设计缓存与并行优化策略，采用 LRU / LFU 混合缓存机制，实现 95%+ hit rate，减少 90% quota usage 与 60% 组件运行时间，并提升 pipeline 鲁棒性。
- 基于 golden set 持续进行 prompt 调优与评测指标优化，提升 relevance evaluation 的 precision / recall 表现，支撑数据标注与迭代闭环。

Multi-modal LLM Pipeline & Data Infrastructure

- 构建多模型内容理解与监控 pipeline，对文本、图像和视频内容进行理解、分类与质量监控，辅助推荐场景中的 feed 质量分析，覆盖 dynamic prompt、cache reuse 与 pipeline 优化等核心环节。
- 基于 ClickHouse 与 Spark 搭建大数据管道，聚合用户、内容与请求信号数据；通过重构连接逻辑与依赖关系提升并行度，将核心 SLA 从 5 天缩短到 4 天，并补充输入输出监控机制以提高数据稳定性。

AI Algorithm Engineer, Huawei

July 2022 – May 2024

负责训练、建模与复杂优化问题求解，侧重 sequence modeling、模型压缩与近似优化，为后续 Agent / LLM 系统工作提供训练与算法基础。

Modeling & Training

- 研究基于 GRU / Transformer 的序列建模与熵编码方案，利用时域概率建模进行 NLP 数据压缩，并结合向量量化压缩模型参数，将编码效率从 6 bits 提升到 5 bits，压缩率提升约 20%。
- 参与端到端深度学习通信系统建模与训练，基于真实相位噪声与功率放大器失真数据设计损失函数并训练神经网络收发机，使传输性能接近 AWGN 理论最优。

Optimization & Approximation

- 针对 NP-hard 资源分配问题，设计随机算法、贪婪策略与选择机制的并行近似求解方案，并结合 bagging / expert 思想进行结果集成，在接近最优整数规划解 95% 的同时，将求解时间从 1500 秒降至 3 秒。

EDUCATION

University of Waterloo

2021-2022

MEng, Electrical & Computer Engineering

Chongqing University

2015-2020

BEng, Electrical Engineering and automation

SKILLS

Languages: Python, Golang, SQL
AI-native Engineering: Problem scoping & prioritization, AI task decomposition, human-in-the-loop workflow redesign, Cursor, Codex
LLM / Agent: Agent harness self-iteration pipeline, auto-evaluation / auto-analysis / auto-iteration, control-theoretic agent loop design (plant / controller / observer / feedback), training-style iteration (benchmark→loss→gradient over context / memory / skill / system prompt), browser-use / computer-use agents, ReAct, context engineering, tool orchestration, system prompt engineering, skill injection & design, LangGraph, RAG, prompt engineering
Training / Modeling: SFT, trajectory data construction, Transformers, GRU, CNN, model compression, offline evaluation, approximation optimization
Data / Infrastructure: Spark, ClickHouse, Azure ML, MySQL, Kafka, CI/CD, RESTful APIs
Language: English, TOEFL 104, GRE 330

PUBLICATION

Transformer-based Fuzzer EAI MONAMI 2022

AWARDS

- 中国大学生数学竞赛：一等奖
- 华为公司 ICT 部门 Top 5 Coder（Python 和 Java）
- 华为"未来之星"